SIMULATING STEADY STATE SPECTRA OF ACROLEIN WHILE ACCOUNTING NUCLEAR QUANTUM EFFECTS

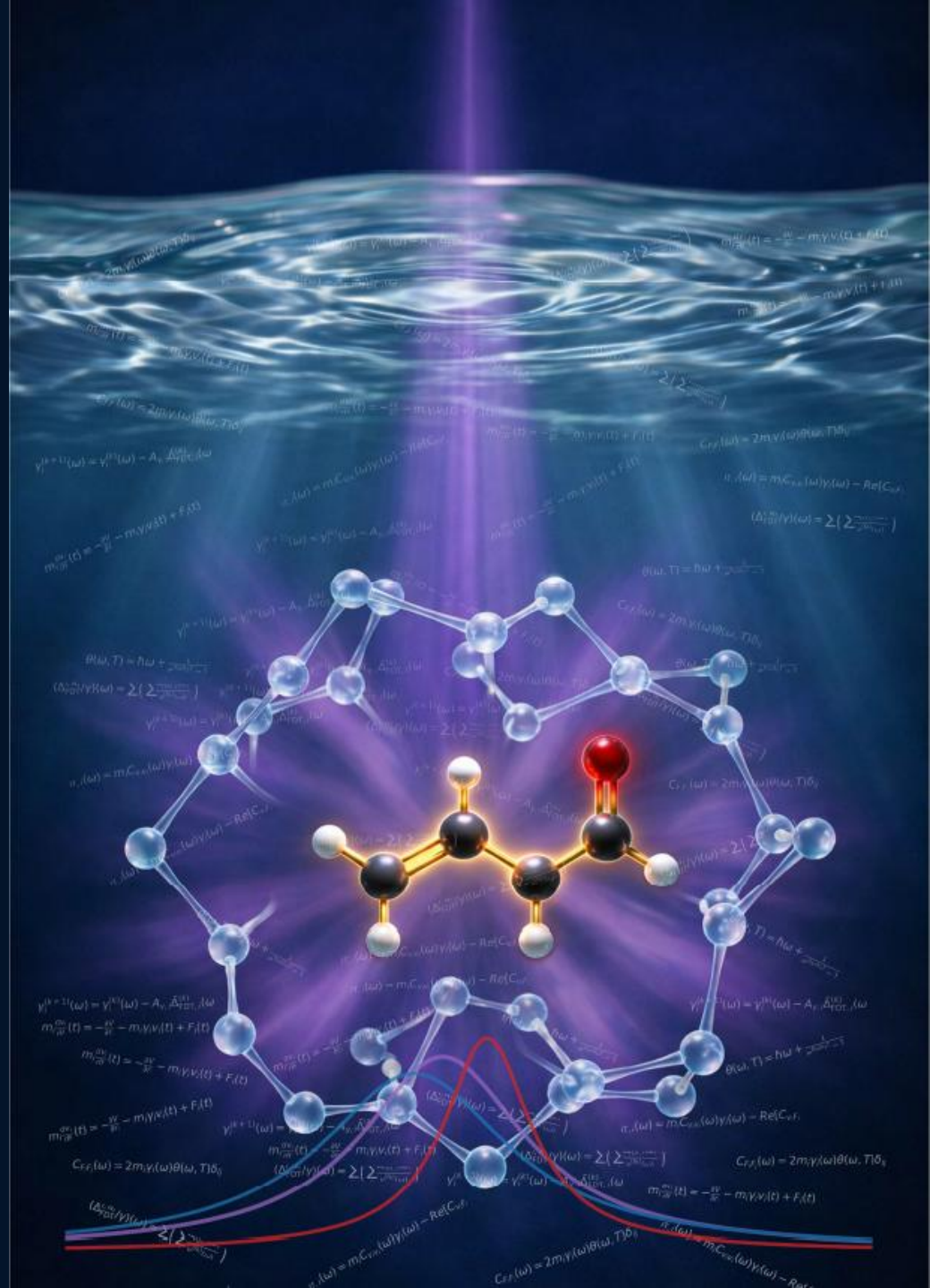
Mukul DHIMAN, Isabelle NAVIZET, Simon HUPPERT and Etienne MANGAUD

J. Phys. Chem. A 2026, 130, 26, 4900–4912

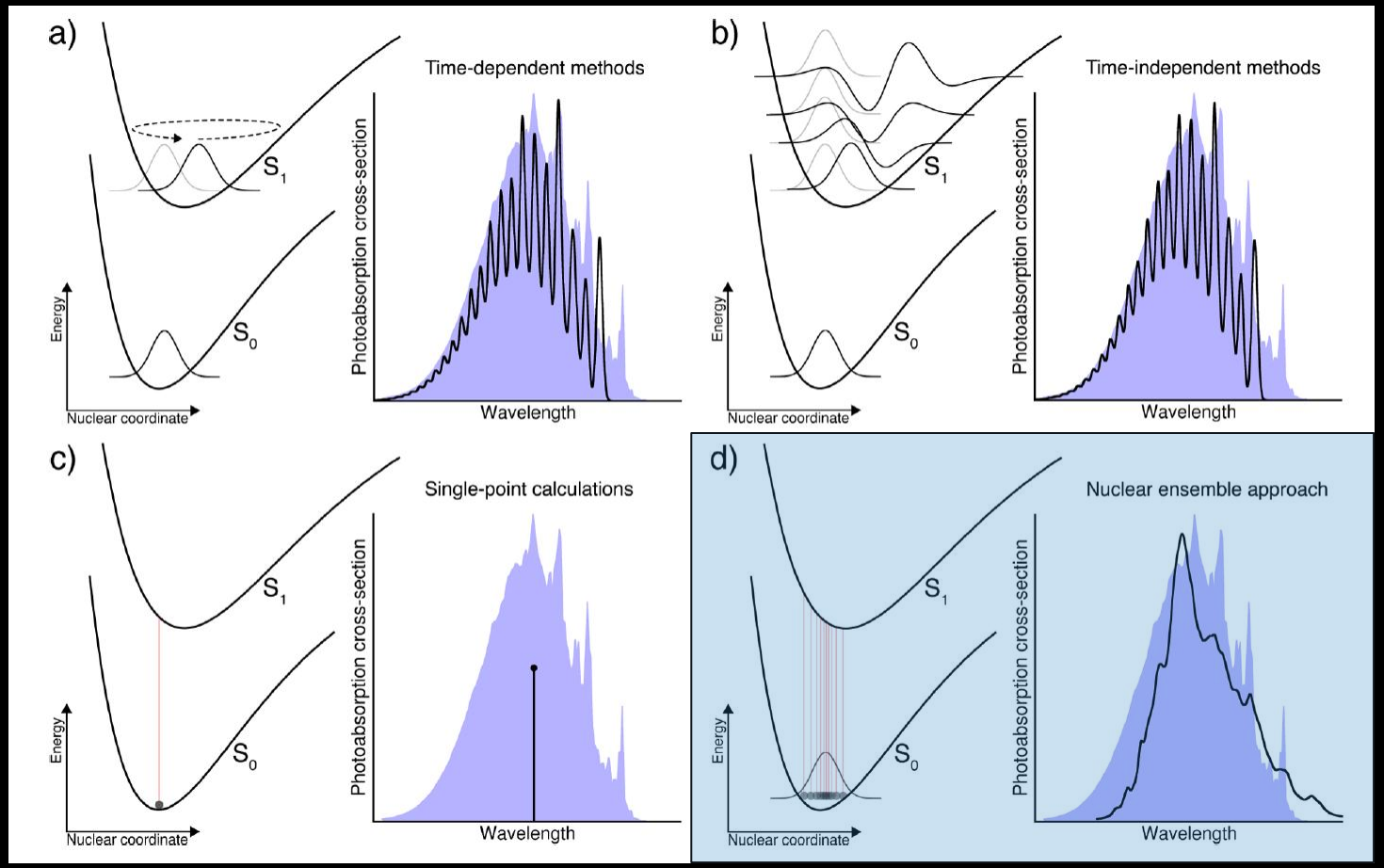
Université Gustave Eiffel

MSME

GDR EMIE Juin 2026



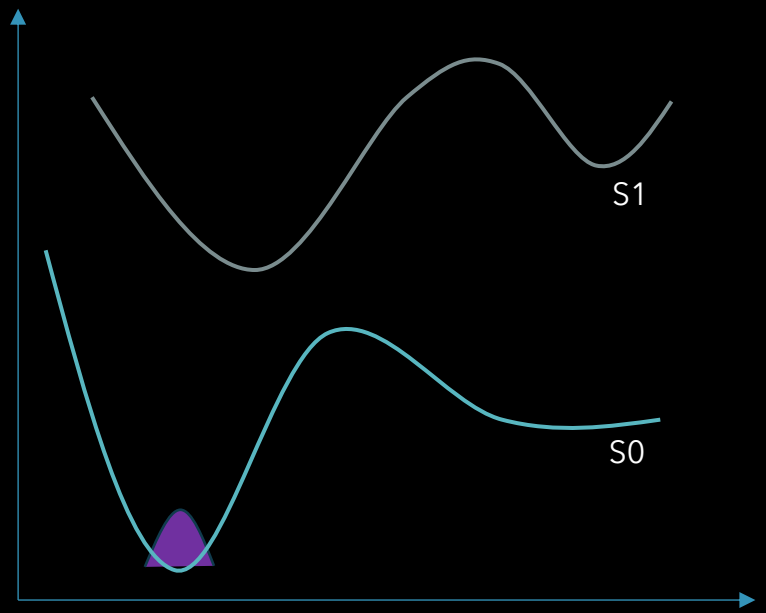
CALCULATION OF PHOTOABSORPTION SPECTRA



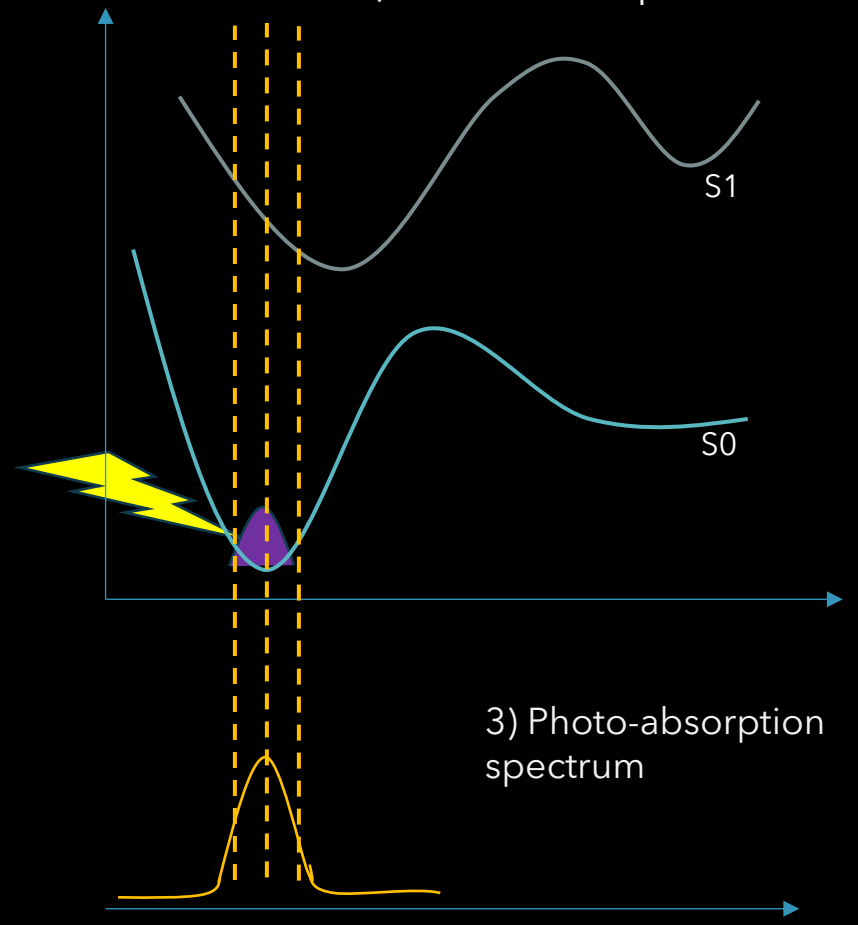
ACS Earth Space Chem. 2022, 6, 207–217

GROUND STATE SAMPLING (NEA)

1) Ground State Preparation



2) Photo-absorption



3) Photo-absorption spectrum

The quality of ground state sampling is directly related the quality of calculated photo-absorption spectra.

GROUND STATE SAMPLING (NEA) METHOD

Langevin (Classical) Dynamics

Each nuclear degree of freedom follows an equation :

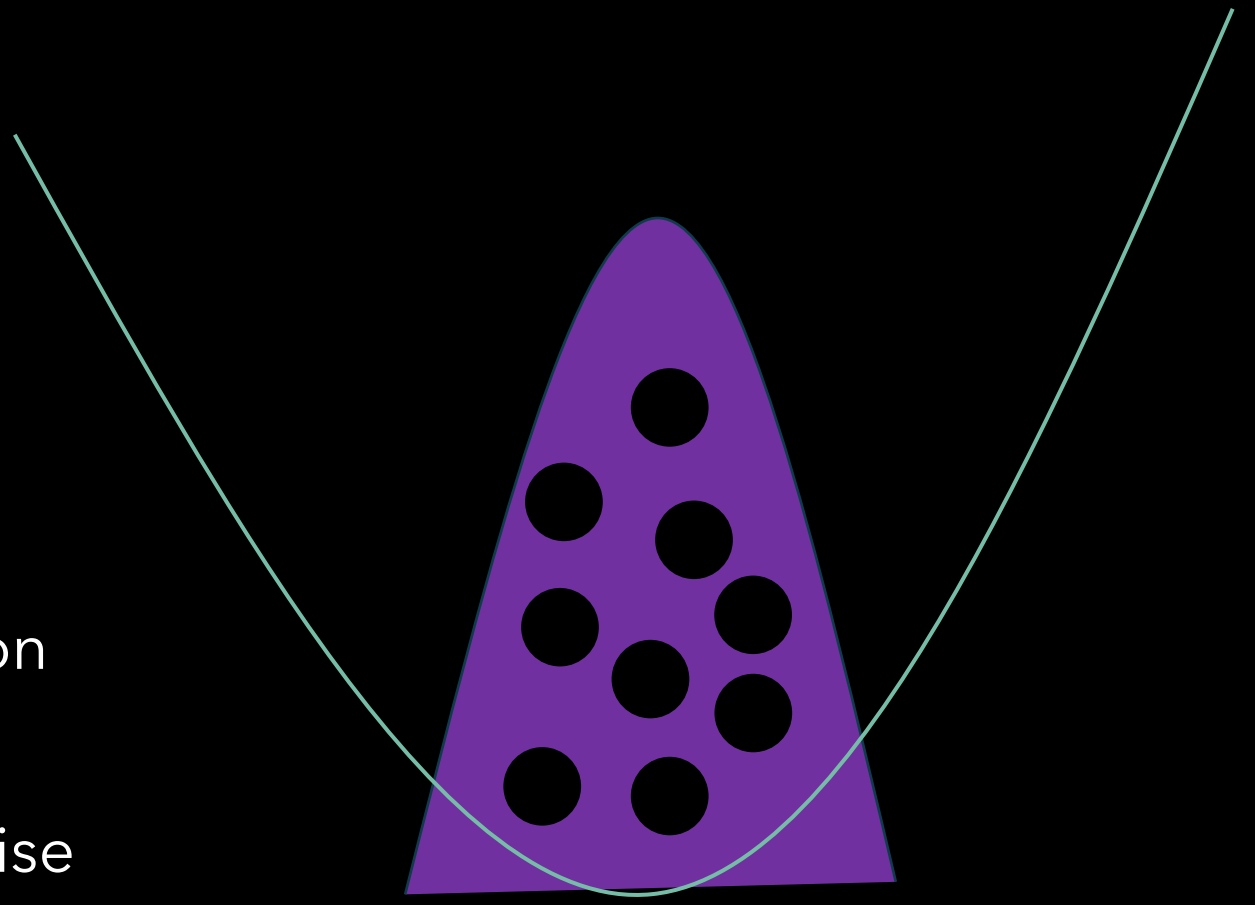
$$m\,\dot{v} = -m\,\gamma\,v - V(\boldsymbol{R}) + F(t)$$

$\gamma \rightarrow$ Friction Term

$V(\boldsymbol{R}) \rightarrow$ Potential energy

$F(t) \rightarrow$ Random Forces with correlation spectrum

$$C_{FF,LGV}(\omega) = 2m\gamma k_B T$$ White Noise



R Kubo 1966 Rep. Prog. Phys. 29 255

DO not describe the zero-point energy

GROUND STATE SAMPLING (NEA) **METHOD**

QTB (Quantum thermal bath) Method

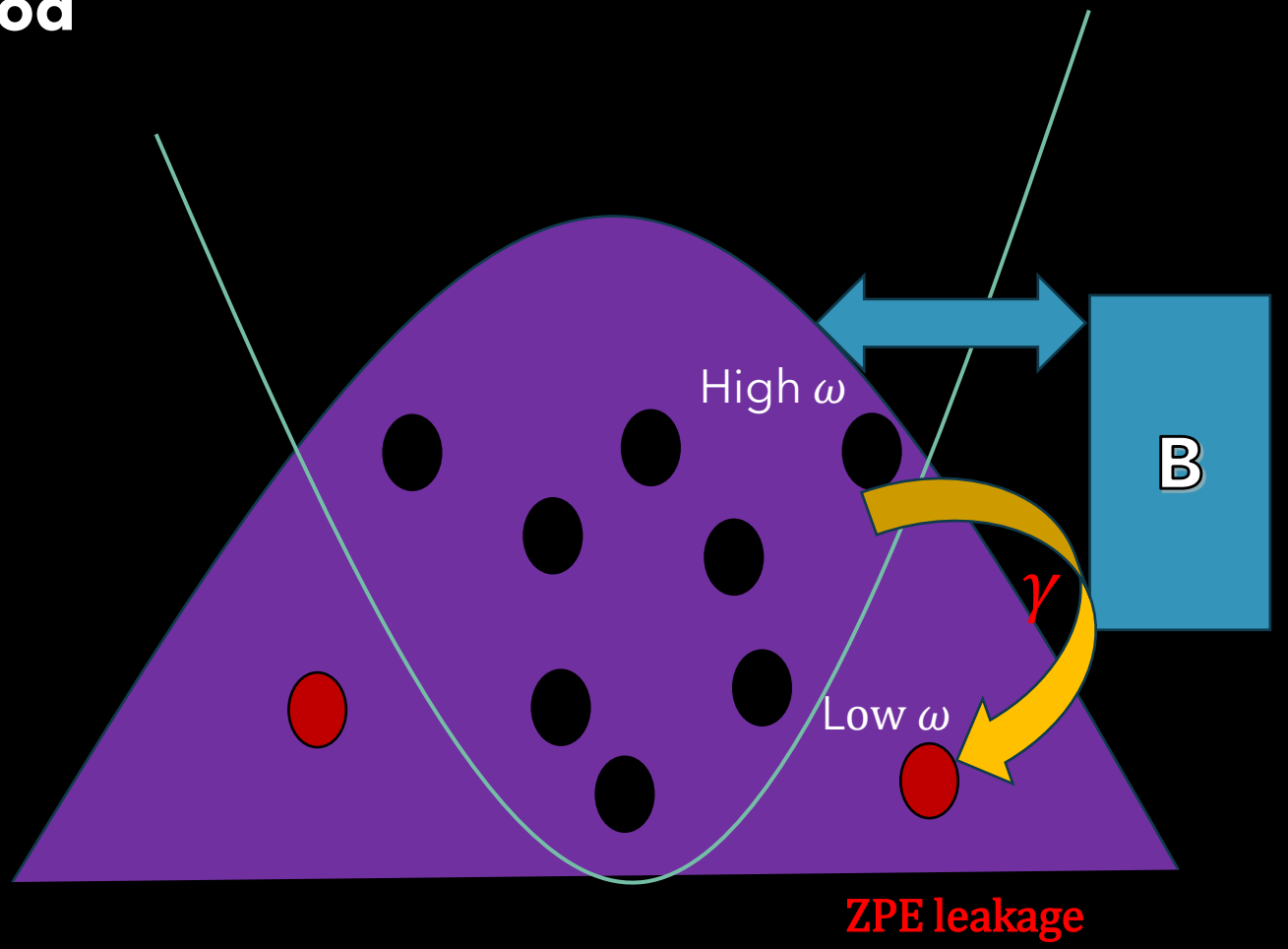
Accounts for the ZPE as for quantum harmonic oscillators

White Noise ⟶ Colored Noise

$$C_{FF,QTB} = 2m\gamma\Theta(\omega,T)$$

With Bose distribution

$$\Theta(\omega,T) = \hbar\omega\left(\frac{1}{2} + \frac{1}{\left(1 - e^{\frac{\hbar\omega}{k_B T}}\right)}\right)$$



Ceriotti *et al* Phys. Rev. Lett. **102**, 020601 (2009)
Dammak et al Phys. Rev. Lett. **103**, 190601 (2009)
Barrat, JL., Rodney, D. *J Stat Phys* **144**, 679–689 (2011)

GROUND STATE SAMPLING (NEA) METHOD

Adaptive - QTB Method

Find ZPE leakage by Quantum fluctuation dissipation theorem

$$\Delta_{FDT}(\omega) = mC_{vv}(\omega)\gamma(\omega) - Re[C_{vF}(\omega)]$$

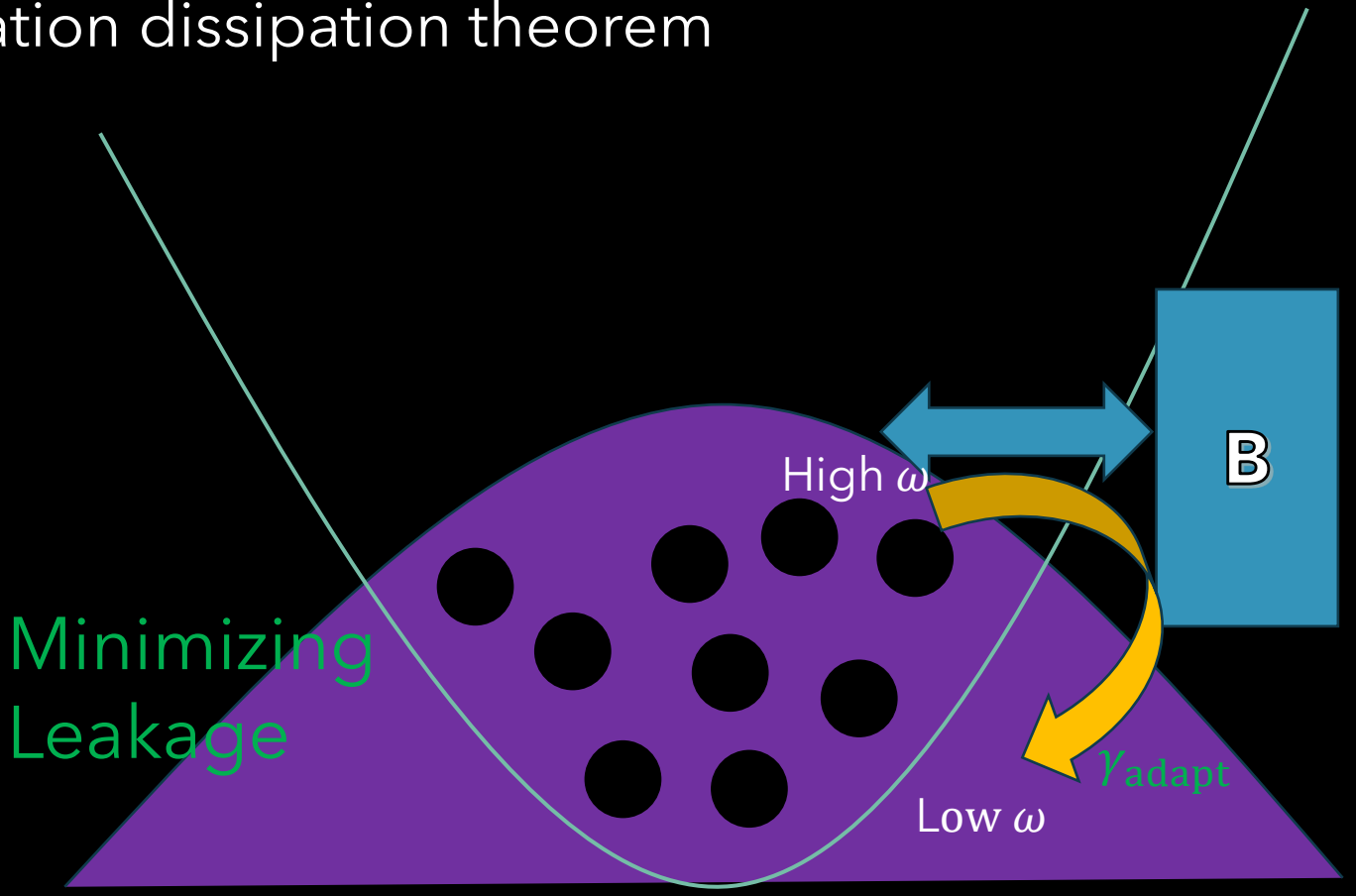
Reduce ZPE leakage by counterbalancing its effects in the correlation spectrum noise

$$C_{FF,adQTB} = 2m\gamma(\omega)\Theta(\omega,T)$$

With an adaptive scheme on a segment of trajectory k

$$\gamma^{k+1}(\omega) = \gamma^{k}(\omega) - A_\gamma \overline{\Delta}^{k}_{FDT}(\omega)$$

$A_\gamma \rightarrow$ adaptation coefficient



Mangaud *at al* J. Chem. Phys.. 2021, 12, 34, 8285-8291
Ple *et al* J. Chem. Theory Comput. 2023, 19, 1432−1445
Mauger at al J. Phys. Chem. Lett. 2021, 12, 34, 8285-8291

GROUND STATE SAMPLING (NEA) METHOD

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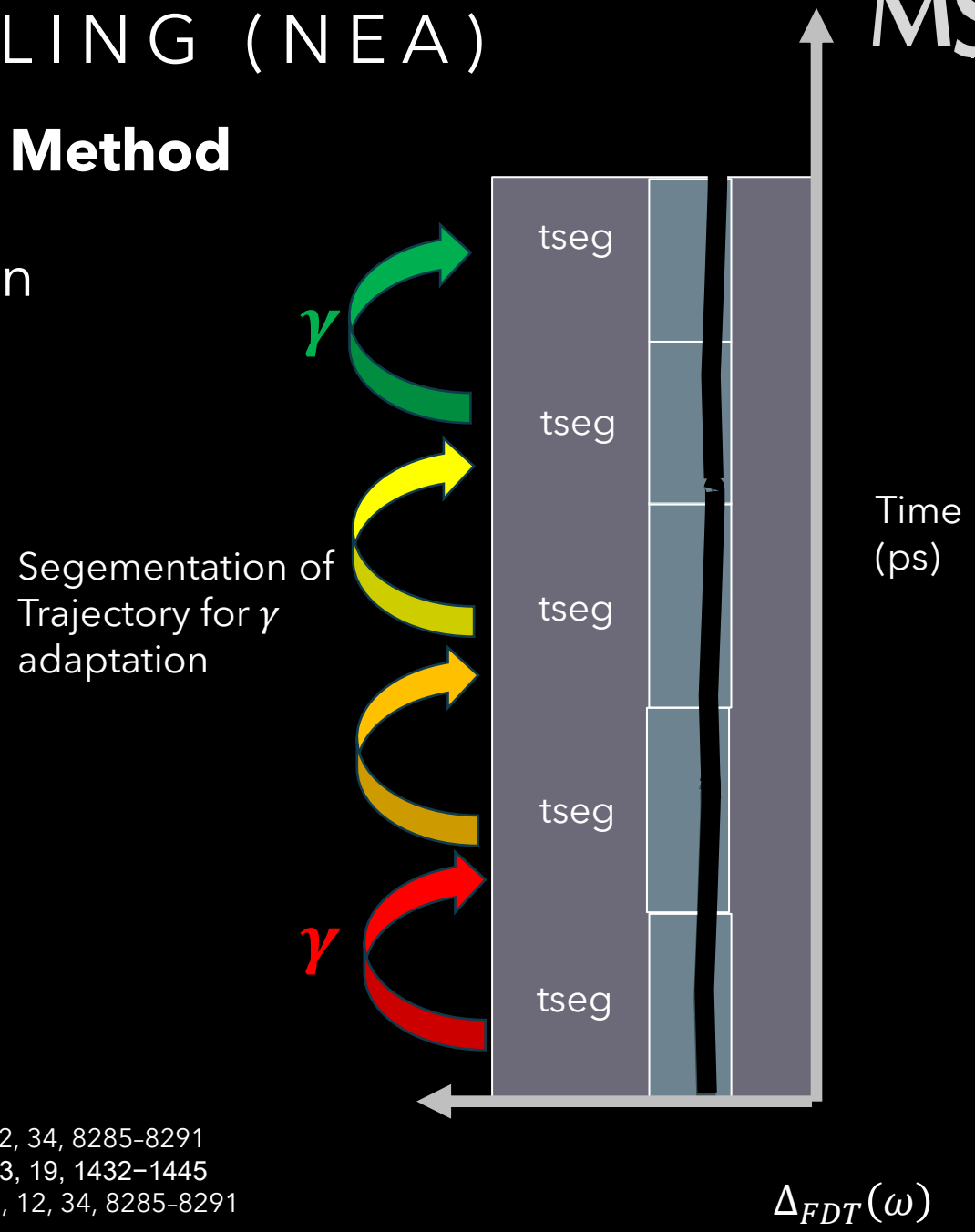
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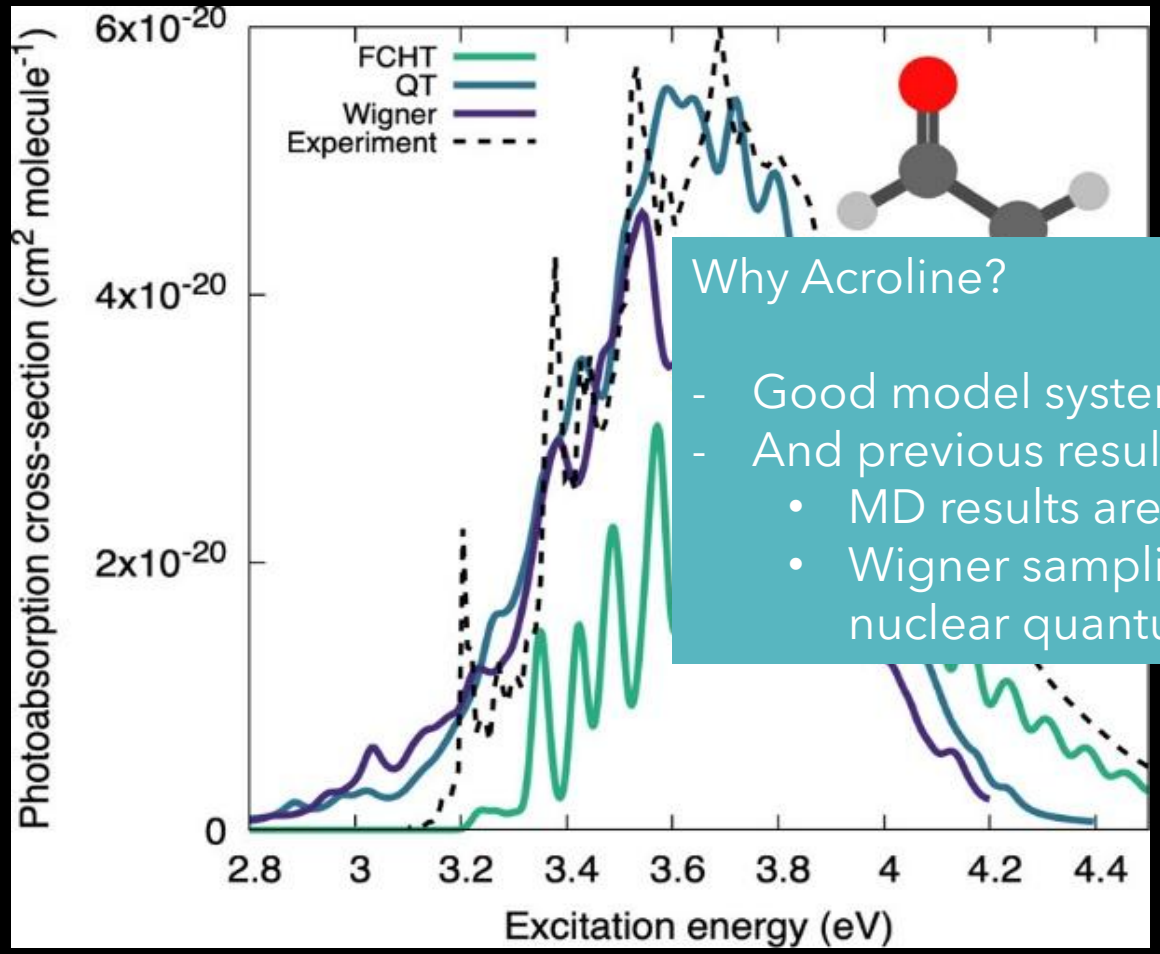
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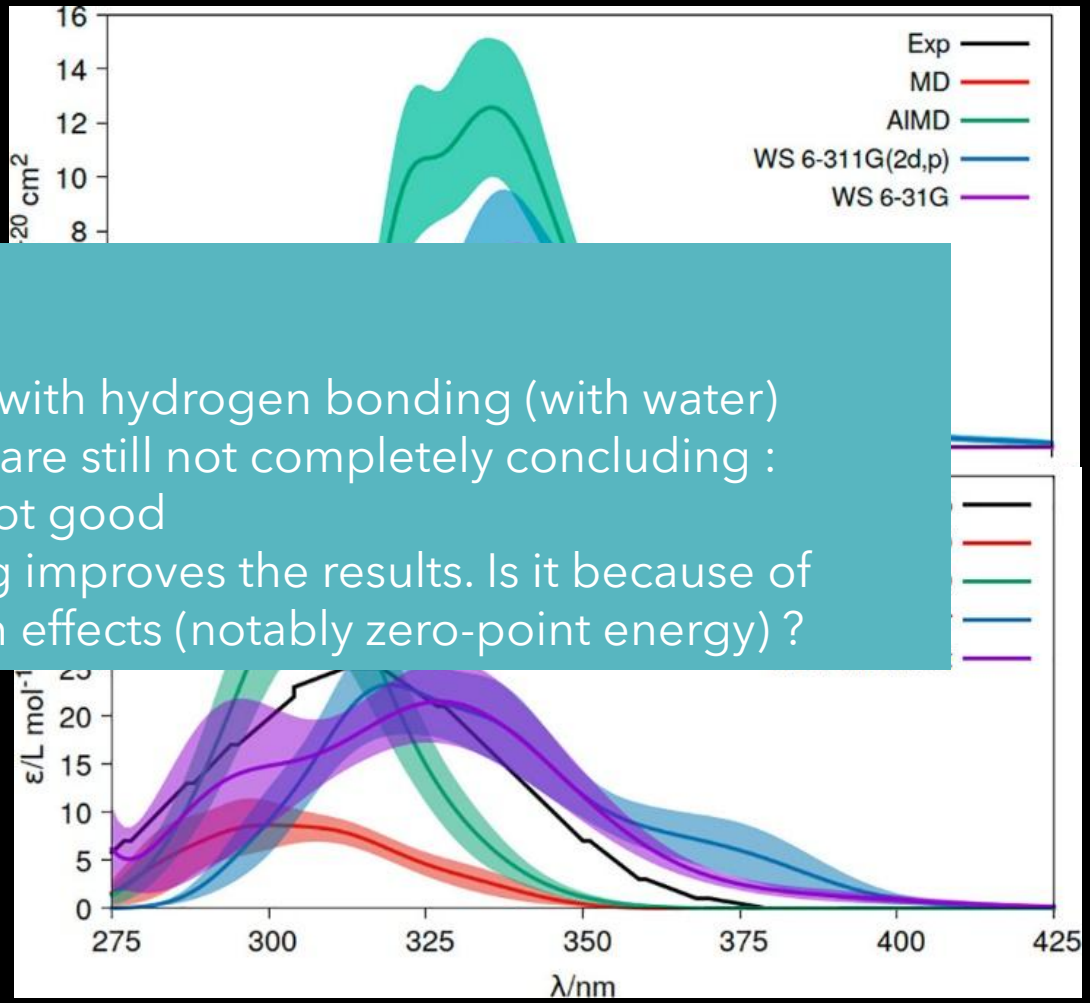
Mangaud *at al* J. Chem. Phys.. 2021, 12, 34, 8285-8291
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Mauger at al J. Phys. Chem. Lett. 2021, 12, 34, 8285-8291

ACROLINE GAS PHASE AND WATER



TD-DFT/TDA/PBE0/6,311G*

A. Prlj ***et al*** ACS Earth Space Chem. 2022, 6, 1, 207–217



A. Borrego-Sánchez *et al* J. Chem. Theory Comput. 2021, 17, 6, 3571–3582

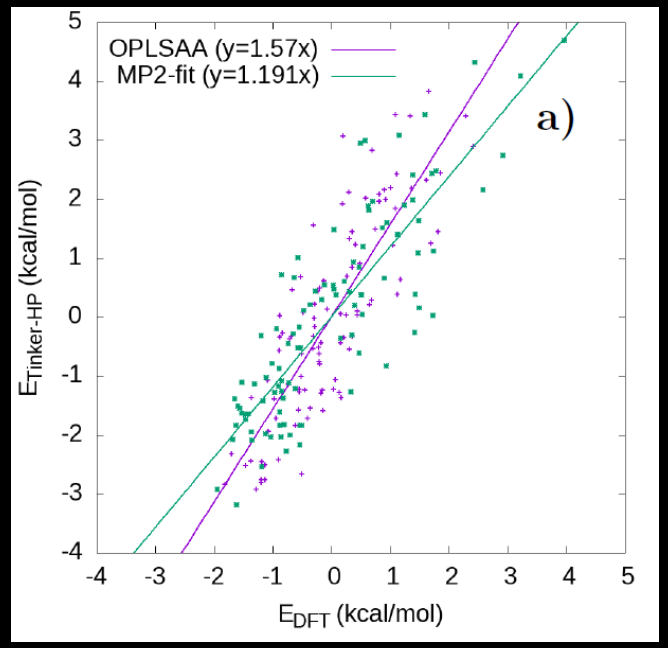
Why Acroline?

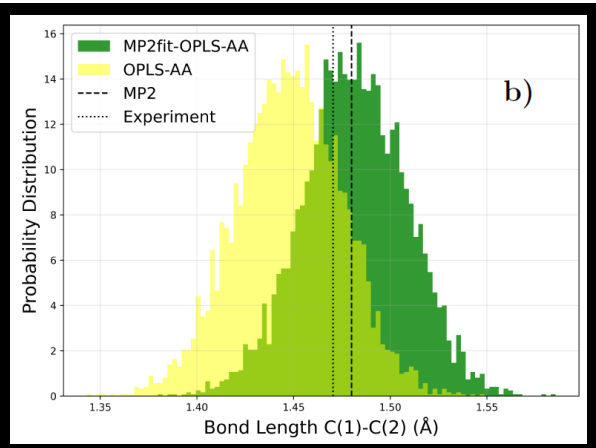
- Good model system with hydrogen bonding (with water)
- And previous results are still not completely concluding :
  - MD results are not good
  - Wigner sampling improves the results. Is it because of nuclear quantum effects (notably zero-point energy) ?

PARAMETRIZATION AND BENCHMARKING

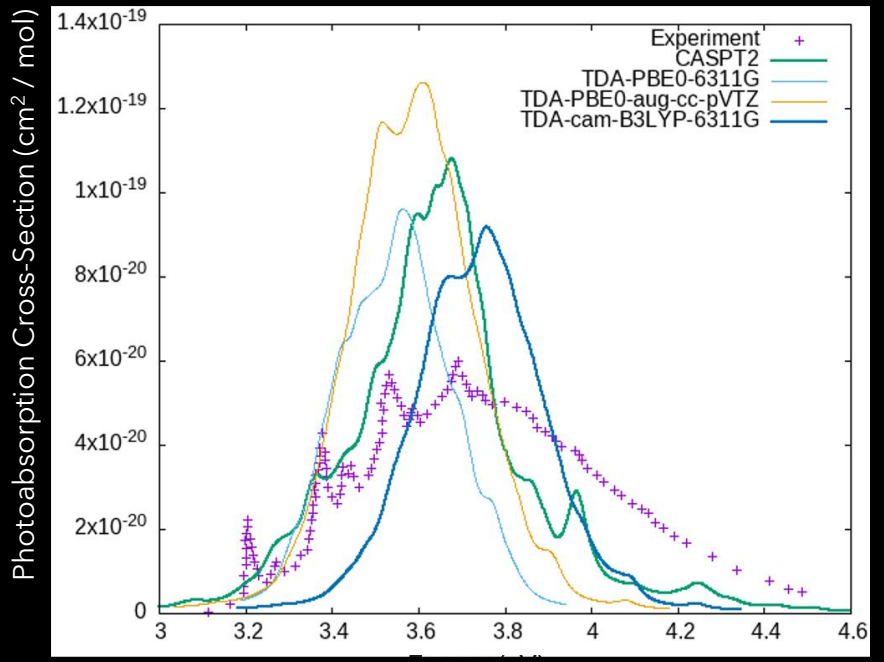
- adQTB is implemented in **Tinker-HP** Software
- Parametrization:
  - Force Fields
  - Electronic Structure method
  - Integrator and Equilibration
  - Selection of friction term (all methods)
  - Selection of (ad)QTB parameters
    - Friction coefficient
    - Segmentation length (tseg)
    - Adaptation Coeff A_γ (ad-QTB)

Force Field Parametrization





DFT Functional and Basis Set

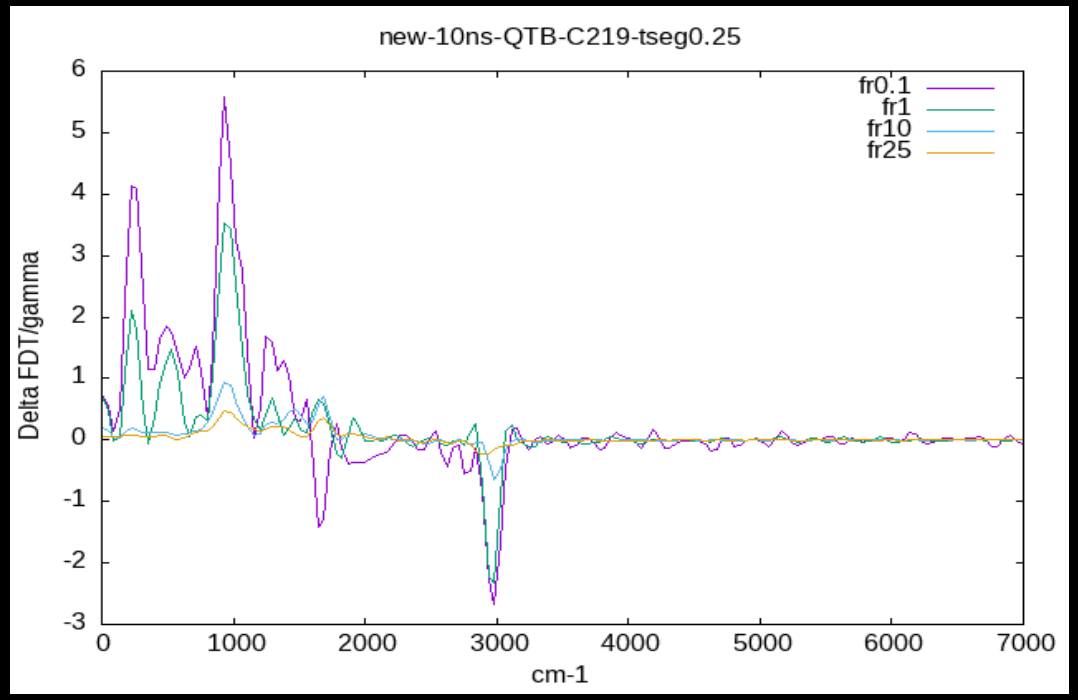


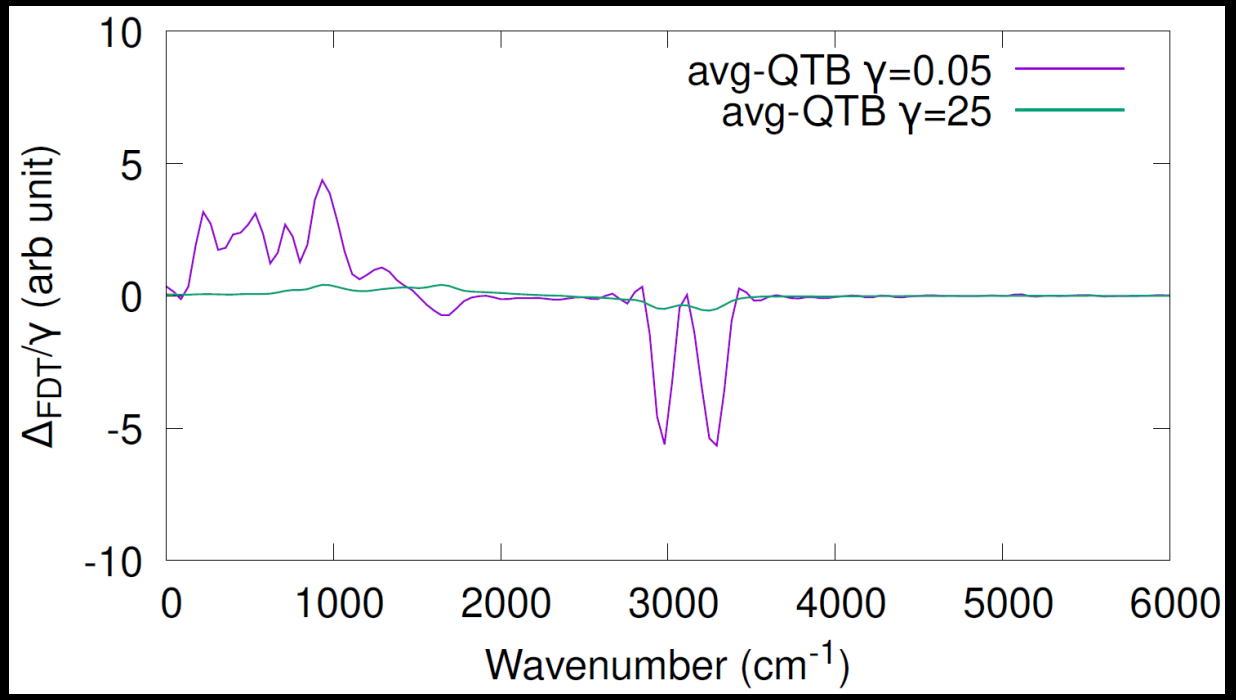
Electronic Structure Methods, TDA-DFT with different functionals compared with CASPT2

TDA-DFT CAM-B3LYP/6311G**

Dhiman *at al* J. Phys. Chem. A 2026, 130, 26, 4900–4912

PARAMETRIZATION AND BENCHMARKING





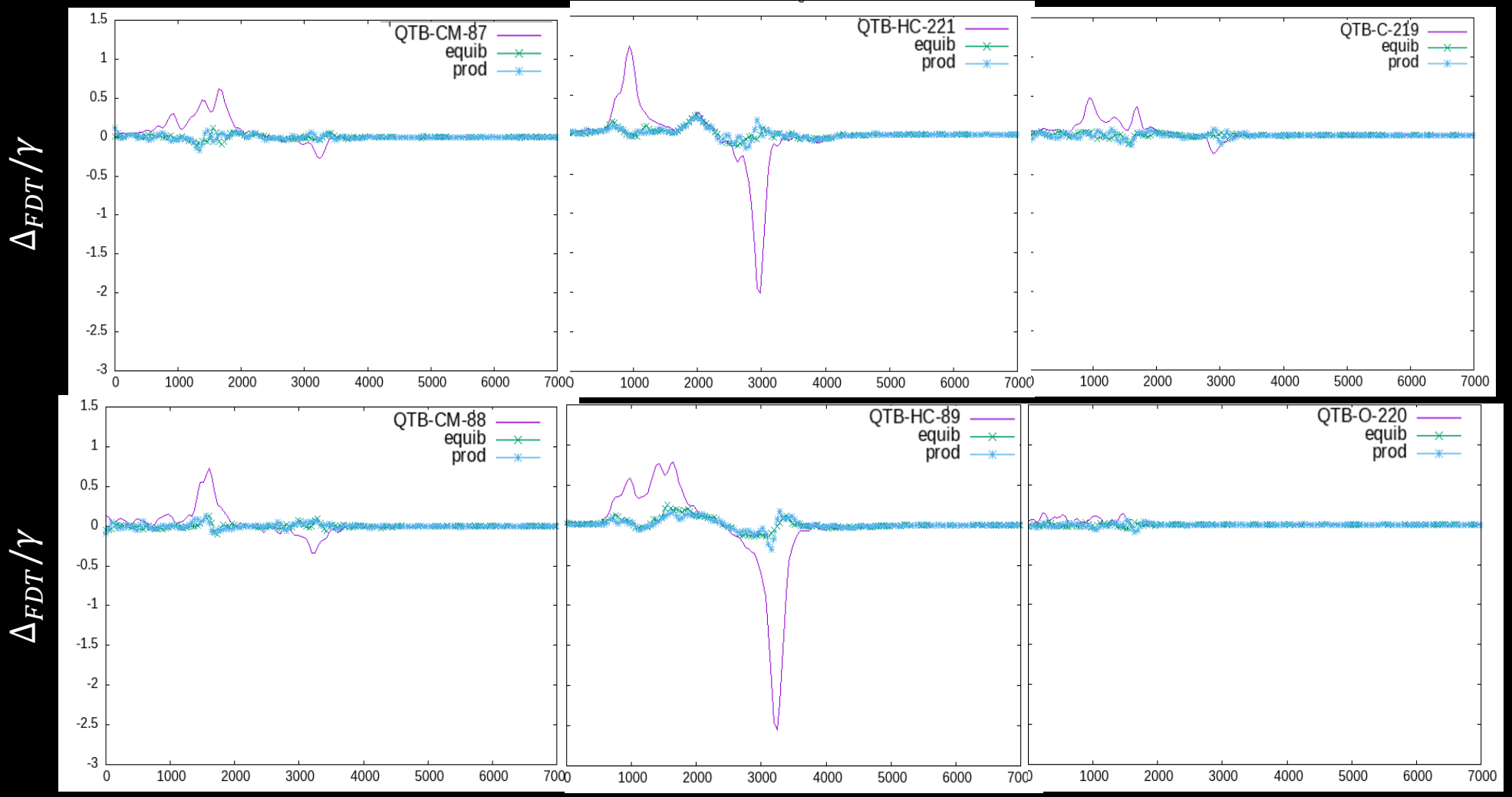
An example of leakage for a mode is shown for different values of friction coefficient (γ) in **QTB** dynamics.

Dhiman *at al* J. Phys. Chem. A 2026, 130, 26, 4900–4912

PARAMETRIZATION AND BENCHMARKING

ad-QTB γ adaptation analysis shows the reduction of leakage for all the modes.

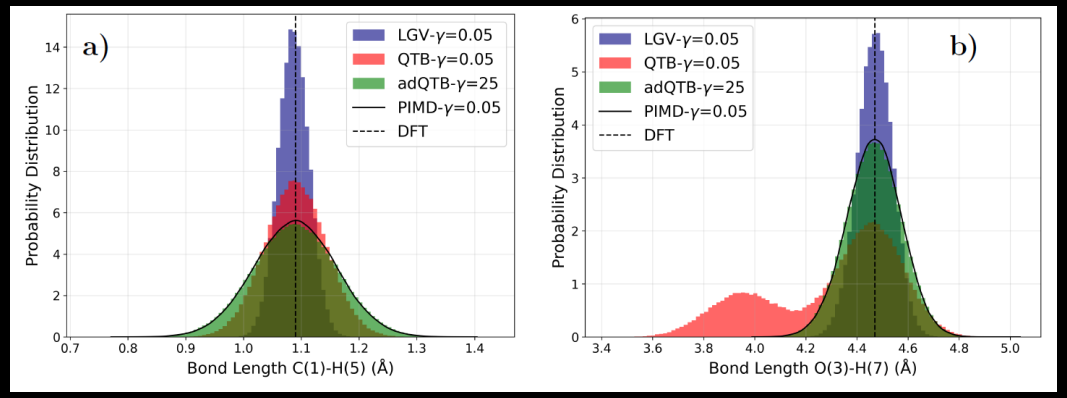
Equilibration time: 40ns
Production: 10ns

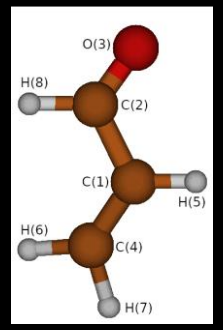


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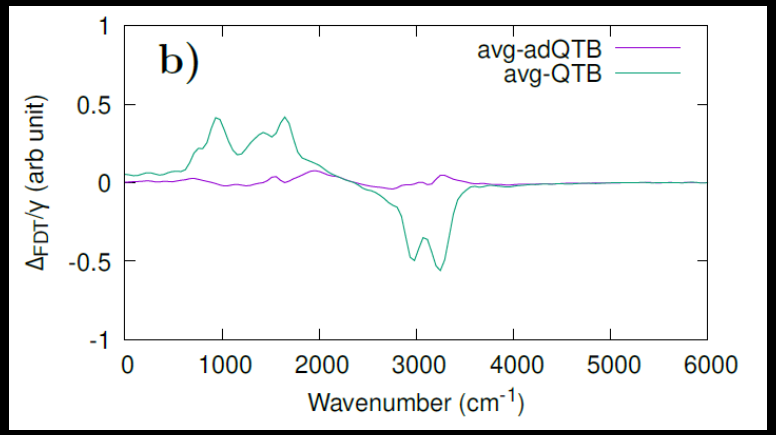
RESULTS: GAS PHASE

Bond Length Distribution Analysis





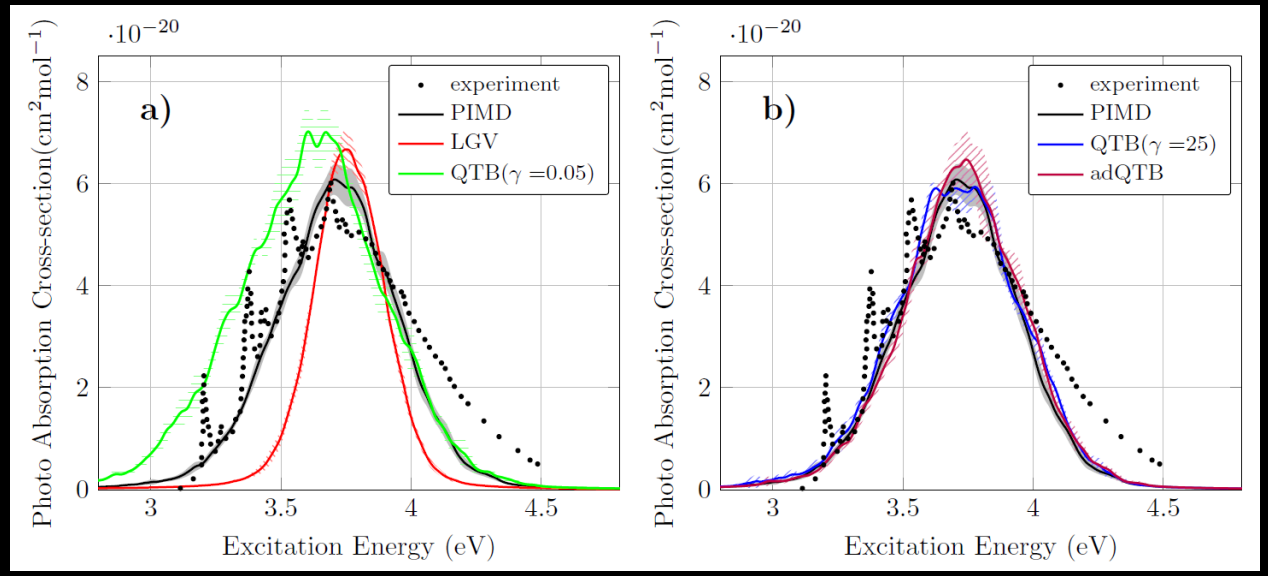
ZPEL Analysis



Photoabsorption Spectrum Analysis

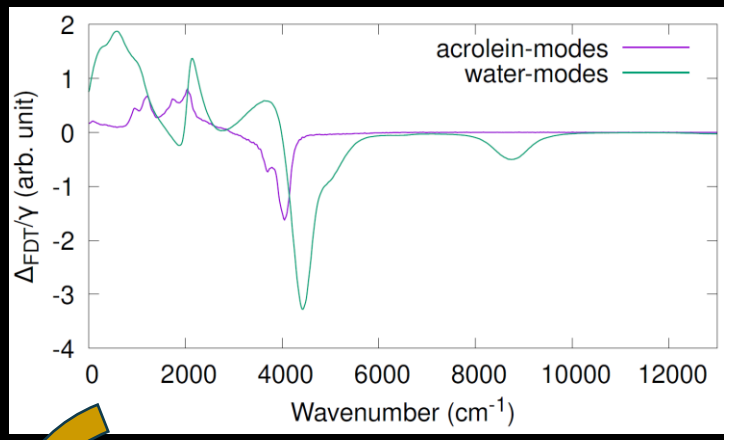
Table 1: Spectral properties for gas phase acrolein computed in this work using LGV, QTB and adQTB method and obtained with experiment[4] and previous work.[1]

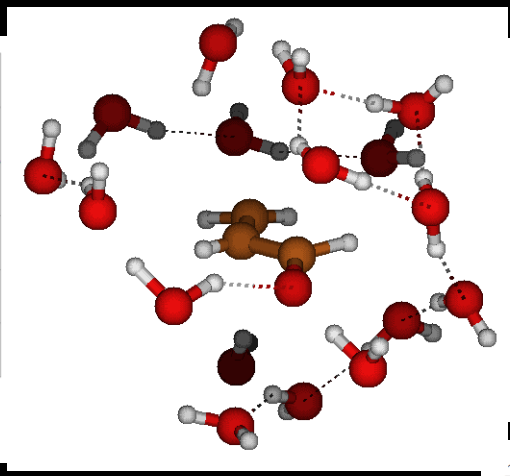
Method	γ ps^{-1}	Position (eV)	Width (FWHM, eV)	Maximum Intensity ($10^{-20}cm^2mol^{-1}$)	Asymmetry Factor Eq:(2)
Experiment[4]	-	3.692	0.654	6.01	0.87
PIMD	0.05	3.707	0.519	6.08	0.86
LGV MD	0.05	3.752	0.309	6.67	0.88
QTB MD	0.05	3.605	0.596	7.02	0.85
QTB MD	25	3.776	0.574	5.94	1.71
adQTB MD	25	3.743	0.505	6.47	1.045
NEA-QT[1]	20	3.74	0.58	6.5	-
NEA–Wigner[1]	20	3.68	0.45	6.2	-



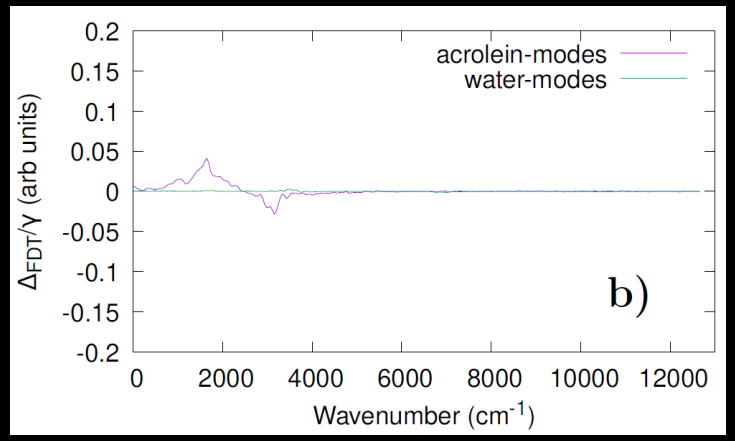
Dhiman *at al* J. Phys. Chem. A 2026, 130, 26, 4900–4912

RESULTS: SOLVENT PHASE





ZPEL Analysis: Post Adaptation

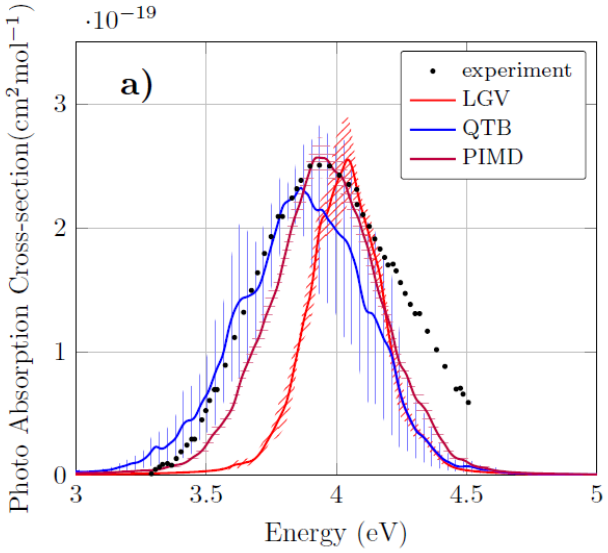


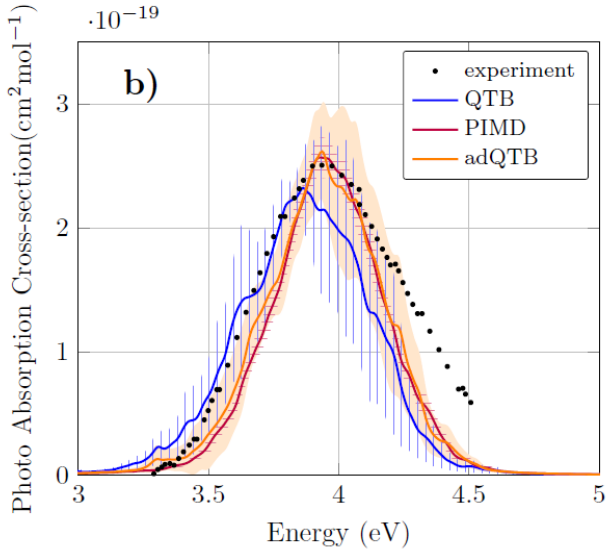
change of scale by factor of 10.

Photoabsorption Spectrum Analysis

Table 2: Spectral Properties Comparison for acrolein in water box (solvent phase).

Method	Position (eV)	Width (FWHM, eV)	Intensity (10^{-19}cm^{2}mol^{-1})	Asymmetry Factor
Experiment	3.935	0.4816	2.505	0.765
LGV MD	4.045	0.2638	2.824	1.080
QTB MD	3.864	0.5655	2.357	1.004
adQTB MD	3.935	0.4940	2.577	0.775
PIMD	3.924	0.4642	2.619	0.715

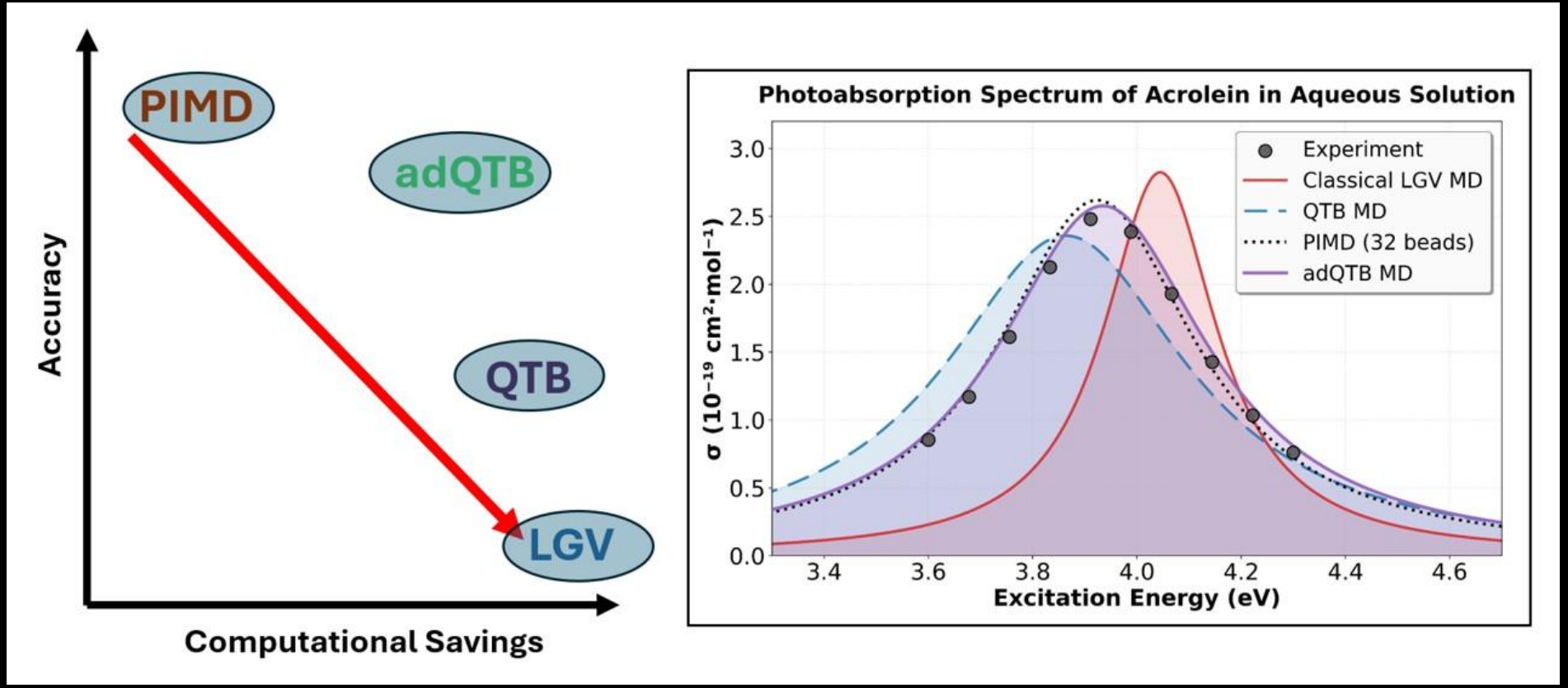




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CONCLUSION

Gas Phase (3.73eV) → Solvent Phase (3.94eV)



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FUTURE WORK

- Results can be improved further by improving the force field parametrisation
- Improve with higher level of electronic structure calculations
- Implement to study varied systems, particularly in solvent phase.
- Study further and find the limit of the solvent effects which can be captured by adQTB
- Further development to study diffusion dynamics on solid surfaces.



Etienne
Mangaud

Isabelle
Navizet

Simon
Hubbart

Thank you !!

